

KAYES, Mali

WMO: 612570

Lat: 14.482N

Lon: 11.399W

Elev: 52

StdP: 100.71

Time Zone: 0.00 (E00)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.0	17.4	-7.6	2.0	26.8	-5.2	2.5	27.8	7.8	28.1	7.3	27.2	2.0	120	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.9	43.7	22.6	42.8	22.4	41.9	22.4	28.1	33.7	27.6	33.2	27.2	32.7	3.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.4	30.6	26.3	21.9	30.3	25.9	21.4	30.1	90.2	33.7	88.2	33.4	86.4	32.8	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.7	6.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			6.7	29.6	1.0	0.7	5.9	30.1	5.3	30.5	4.7	31.0	4.0	31.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.4	25.4	28.2	32.2	34.9	36.5	34.3	29.7	28.6	28.9	30.5	28.9	26.4
	DBStd	3.87	2.03	2.51	2.47	1.98	1.75	2.46	2.02	1.78	1.46	1.47	1.63	2.38
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7436	477	508	687	747	821	729	612	577	567	635	568	508
	CDD18.3	4394	218	275	429	497	563	479	353	318	317	377	318	250
	CDH23.3	62923	2483	3587	6663	8426	9711	7767	4694	3779	3797	5008	4091	2917
	CDH26.7	39121	1259	2084	4431	6059	7238	5398	2432	1663	1758	2803	2446	1549
Wind	WSAvg	3.1	3.2	3.4	3.0	3.1	3.5	4.0	3.5	2.8	2.5	2.4	2.4	3.1
Precipitation	PrecAvg	629	0	0	0	0	12	73	152	221	133	41	1	0
	PrecMax	890	3	14	4	6	40	170	340	407	268	203	20	9
	PrecMin	428	0	0	0	0	0	0	5	80	59	0	0	0
	PrecStd	116	1	2	1	1	11	39	70	82	49	36	3	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	40.8	43.2	44.6	45.0	43.5	38.6	37.0	36.0	39.4	39.5	38.7
		MCWB	18.1	20.3	20.9	23.2	22.9	23.6	26.4	26.0	27.1	23.3	20.8	20.0
	2%	DB	35.3	39.0	41.8	43.7	43.9	42.0	36.7	35.0	34.8	38.0	38.5	37.0
		MCWB	17.6	19.5	20.8	22.5	22.8	23.9	26.3	26.6	27.2	24.1	20.7	19.2
	5%	DB	34.1	37.5	40.7	42.8	42.9	40.7	35.4	33.6	34.0	37.0	37.5	35.6
		MCWB	17.1	18.8	20.4	22.0	22.8	24.3	26.4	26.8	27.3	25.2	20.6	18.4
	10%	DB	32.9	36.0	39.6	41.7	42.0	39.6	34.3	32.6	33.1	36.0	36.4	34.0
		MCWB	16.4	18.0	19.9	21.7	23.0	24.5	26.3	26.6	27.0	25.3	20.4	17.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.8	24.3	26.3	26.6	27.4	27.7	28.2	28.7	28.5	25.6	22.0
		MCDB	32.7	37.6	40.1	41.8	39.5	36.7	35.0	32.7	33.2	34.7	31.6	35.6
	2%	WB	19.1	20.8	22.8	24.7	26.0	26.7	27.2	27.7	28.1	27.8	24.4	20.5
		MCDB	33.1	36.9	38.3	39.6	38.3	35.8	33.7	32.2	33.0	34.3	31.1	34.1
	5%	WB	18.0	19.7	21.6	23.9	25.4	26.2	26.9	27.2	27.7	27.3	23.0	19.2
		MCDB	31.8	35.7	38.1	38.6	37.6	35.1	33.0	31.7	32.5	33.6	31.8	33.3
	10%	WB	17.1	18.7	20.7	23.1	24.9	25.8	26.6	26.9	27.3	26.9	21.8	18.3
		MCDB	30.9	34.4	37.8	38.8	37.0	34.7	32.4	31.3	31.7	33.0	33.3	32.7
Mean Daily Temperature Range	5% DB	MDBR	15.1	14.9	14.8	13.6	10.9	9.7	7.9	7.3	8.3	11.7	16.3	15.2
		MCDBR	17.0	16.7	15.7	14.4	12.3	10.8	9.2	8.5	9.3	13.0	17.7	17.2
		MCWBR	7.1	6.5	6.0	5.3	4.5	3.0	2.6	2.9	3.4	3.9	5.8	6.4
	5% WB	MCDBR	16.0	16.2	14.2	12.8	10.1	9.8	8.1	7.9	8.7	11.0	15.3	15.8
		MCWBR	7.1	6.7	6.0	5.4	3.7	3.0	2.9	3.4	3.8	4.1	5.7	6.3
Clear-Sky Solar Irradiance	taub	0.515	0.552	0.656	0.662	0.699	0.727	0.653	0.584	0.576	0.596	0.511	0.485	
	taud	1.896	1.806	1.600	1.634	1.586	1.598	1.763	1.924	1.919	1.855	1.967	1.997	
	Ebn at Noon	777	769	702	698	661	635	687	745	752	724	775	792	
	Edh at Noon	191	217	273	264	273	266	226	195	196	204	176	168	
All-Sky Solar Radiation	RadAvg	5.33	5.95	6.61	6.96	6.76	6.26	5.77	5.44	5.78	5.82	5.43	5.17	
	RadStd	0.33	0.30	0.31	0.26	0.28	0.23	0.29	0.24	0.24	0.12	0.23	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page